Regulation & Pharmacology of the Hypothalamus Pituitary Axis

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Block: Endo/Uro/Repro
Thread: Physiology / Pharmacology
Conflict of Interest: Dr. Haines has no conflicts of interest to disclose

Learning Objectives

By exam time, you should be able to do the following:

1. Recall the hormones, target organs, functions and feedback regulation of the hypothalamic-pituitary-growth hormone axis – our model for understanding the system. PHYS
2. Explain the pharmacologic treatment of growth hormone excess and deficiency, including the mechanisms of action, key adverse effects, and monitoring requirements of relevant medications. PHARM
3. Describe the hormones, target organs, functions, and feedback regulation of the hypothalamic-pituitary-adrenal axis. PHYS
4. Explain the pharmacologic treatment of adrenal hormone excess (primary and secondary) and deficiency (primary and secondary), including the mechanisms of action, key adverse effects, and monitoring of agents used in the medical management of these disorders. PHARM
5. Describe the hormones, target organs, functions, and feedback regulation of the hypothalamic-pituitary-thyroid axis. PHYS
6. Explain the pharmacologic treatment of thyroid hormone excess (primary and secondary) and deficiency (primary and secondary), including the mechanisms of action, key adverse effects, and monitoring of agents used in the medical management of these disorders. PHARM
7. Describe the hormones, target organs, functions, and feedback regulation of the hypothalamic-pituitary-gonadal axis and hypothalamic-pituitary-breast axis. PHYS
8. Explain the pharmacologic treatment of prolactinomas, including the mechanisms of action, key adverse effects, and monitoring of agents used in the medical management of these disorders PHARM.
9. Describe the action and release of the posterior pituitary hormones oxytocin and anti-diuretic hormone. PHYS
10. Explain the pharmacological treatment of ADH deficiency and central diabetes insipidus, including the mechanisms of action, key adverse effects, and monitoring of agents used in the medical management of these disorders. PHARM

Study Questions

1. How do end-organ hormones feedback on the components of the hypothalamic-pituitary-end organ axis in most cases?

2. What is unique about growth hormone when compared to all other hormones produced by the anterior pituitary?

3. Why does ACTH overproduction, when seen in Addison's disease, lead to skin hyperpigmentation?

4. Why are serum TSH measurements generally reliable, but serum ACTH measurements often are not?

5. What is the product of Sertoli cells? How does that product lead to negative feedback of the H-P-gonadal axis in persons with testes?

6. What are the main components of the H-P-gonadal axis in persons with ovaries? How does this axis function?

7. What is unique about the H-P-breast axis when compared to other H-P-end organ axes?

8. How does oxytocin act and why does it become so potent at the time of labor?

9. A 50-year-old patient has a large pituitary adenoma. You suspect that he may have suboptimal cortisol production due to ACTH deficiency. His AM cortisol is in the low normal range. What is the best test to confirm your suspicion?
    a. Cosyntropin stimulation test
    b. Repeat AM cortisol
    c. 24-hour urinary free cortisol
    d. 1 mg dexamethasone suppression test

10. A 28-year-old with central hypothyroidism is placed on levothyroxine (synthetic thyroxine). What test should you measure to ensure appropriate replacement?
    a. TSH
    b. TSH, free t4 and free t3
    c. Free t3
    d. Free t4

11. A patient with pituitary Cushing's is unable to undergo transsphenoidal resection due to medical reasons. You select an agent that decreases ACTH secretion. That agent is:
    a. Pasireotide
    b. Mifepristone
    c. Ketoconazole
    d. Metyrapone

12. A patient has a TSH secreting adenoma and needs medical therapy. The best agent to use is:
    a. Methimazole
    b. Cabergoline
    c. Prednisone
    d. Octreotide

13. A patient is diagnosed with central diabetes insipidus. You place them on desmopressin. The most common and dangerous side effect is:
    a. Anaphylaxis
    b. Hyponatremia
    c. Hypernatremia
    d. Heart failure

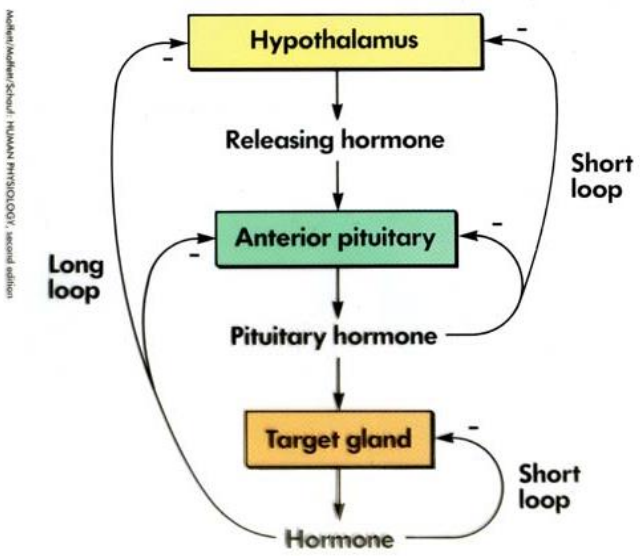
A. Basic schematic of Hypothalamic-Pituitary-end organ axis

1. The production of pituitary tropic hormones is under direct regulation by hypothalamic neurohormones. The release of anterior pituitary hormones is cyclic. This pattern of hormone release is thus governed by the nervous system. Both sleep and circadian patterns produce a rhythmic release of anterior pituitary hormones.

2. Production of anterior pituitary hormones negatively feeds back on the hypothalamus, decreasing production and release of hypothalamic hormones. Anterior pituitary hormones also cause direct negative feedback on the trophic cells of the anterior pituitary.

3. Anterior pituitary hormones enter the general circulation and stimulate the production and release of hormones at the end organ. These end organ hormones negatively feed back to the hypothalamus, the pituitary, and to the end organ itself, mitigating production and release of their particular hormones.

Figure 1



B. Hypothalamic and pituitary hormones summarized

1. Trophic cell types in the anterior pituitary are subject to hypothalamic control with stimulatory hormones (like TRH and CRH) and inhibitory hormones (like dopamine and somatostatin).

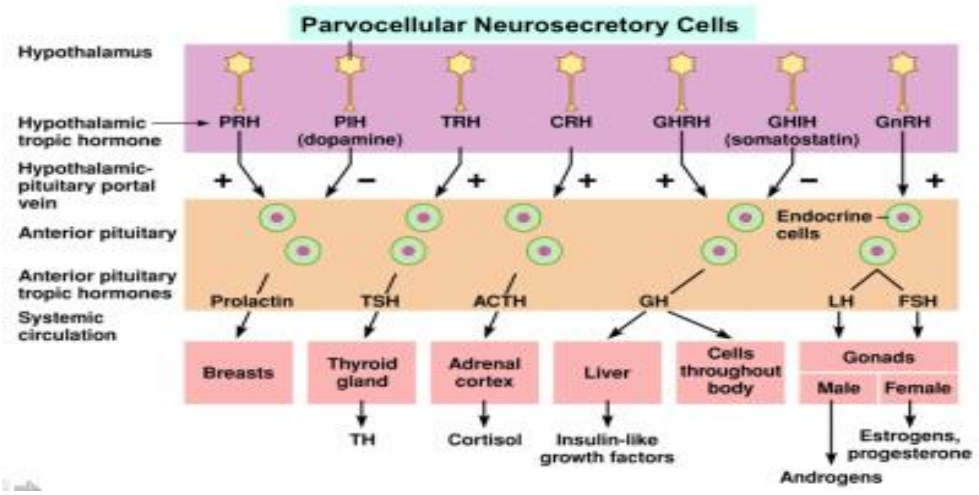


Figure 2

C. Structure of anterior pituitary

1. Peptide hormones from the hypothalamus enter the hypothalamic-anterior pituitary portal circulation and bathe the tropic cells of the anterior pituitary in high concentrations of stimulatory and inhibitory hormones.

2. Extracellular receptors on trophic cells bind stimulatory hormones and cause transcription, storage and release of anterior pituitary peptide hormones. These peptide hormones are released into the systemic circulation.

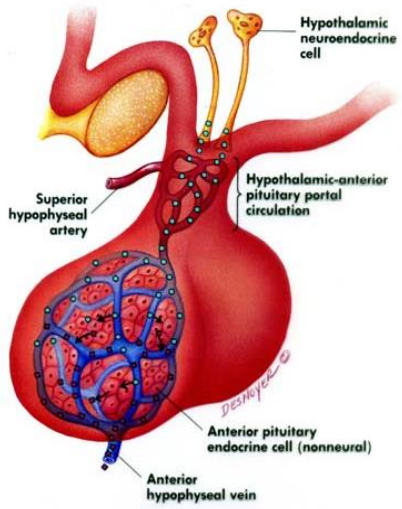


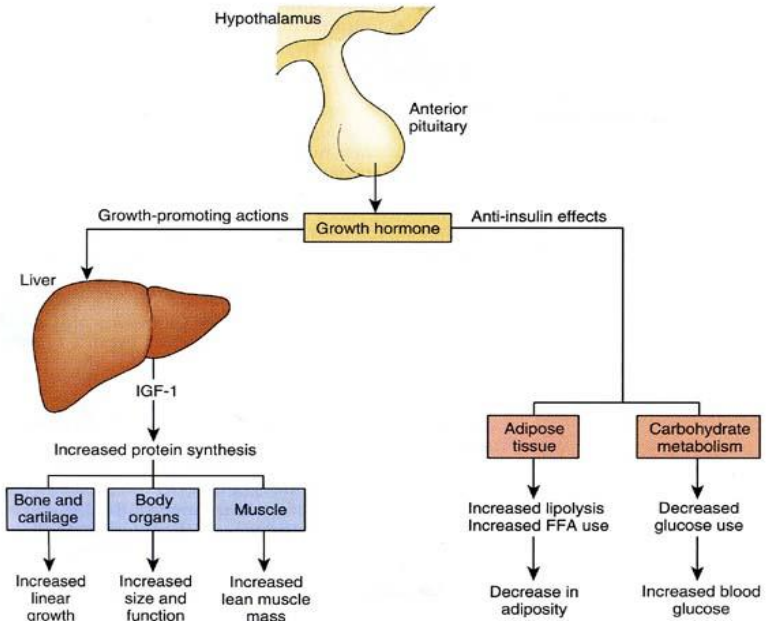
Figure 3

D. H-P-Growth Hormone axis revisited

1. Growth hormone is the only anterior pituitary hormone with significant effects systemically and on a specific end organ.

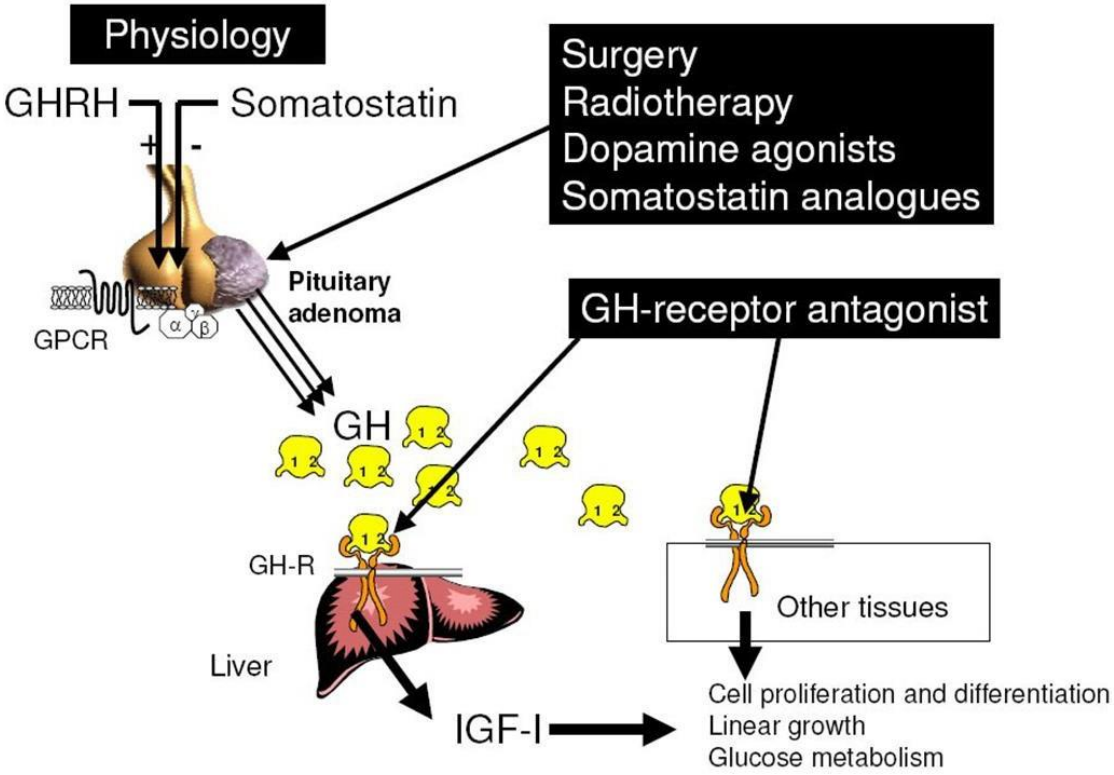
2. GHRH, GH, and IGFs all negatively feed back on the hypothalamus. GH and IGFs negatively feed back on the somatotrophs of the anterior pituitary. IGFs negatively feeds back on the liver.

Figure 4



3. Pathophysiology and pharmacology

   a. Excess of growth hormone. As seen in our previous lecture, growth hormone excess (usually from an adenoma of somatotropes) causes gigantism before puberty and acromegaly afterwards. The goals of therapy in patients with acromegaly are to lower the serum insulin-like growth factor-1 (IGF-1) concentration to within the normal range for the patient's age and



gender, control adenoma size and reduce mass effects, improve symptoms, and reverse metabolic abnormalities such as diabetes mellitus.

Somatostatin is a key regulatory peptide that functions primarily as a paracrine mediator. The physiological effects of somatostatin are largely inhibitory. Somatostatin analogs were developed that are much more stable in the circulation and are more potent in many of the inhibitory actions than native somatostatin. **Octreotide**, **lanreotide**, and **pasireotide** are somatostatin analogues used to treat acromegaly.

Pegvisomant is a GH receptor antagonist that blocks native GH from binding. Liver enzyme monitoring is necessary.

b. Deficiency of growth hormone. Growth hormone deficiency (GHD) is effectively excluded in children with a normal bone age and height velocity.

IGF-I is the mediator of the majority of the growth-promoting actions of GH. Patients with suspected GHD should be screened with levels of insulin-like growth factor I (IGF-I). GH levels can vary markedly within a normal day, thus IGF-1 remains the test of choice.

Recombinant human GH (rhGH) is administered by subcutaneous injection once a day, usually in the evening. Measurement of serum IGF-1 is the best single test of the adequacy of GH treatment. In pediatric patients, monitoring growth response (i.e., height or height velocity) is also important. The most common side effects of GH treatment in patients with hypopituitarism are peripheral edema, headaches, arthralgias, carpal tunnel syndrome, paresthesia, and worsening of glucose tolerance.

E. Structure of H-P-adrenal axis

1. Corticotropin releasing hormone is released by the hypothalamus. This bathes the corticotropes of the anterior pituitary and leads to the release of adrenocorticotropic hormone (ACTH). ACTH enters the general circulation and binds to extracellular receptors on the cells of the adrenal cortex. In response, the cells of the adrenal cortex produce and release cortisol, androgens, and to a lesser extent aldosterone.

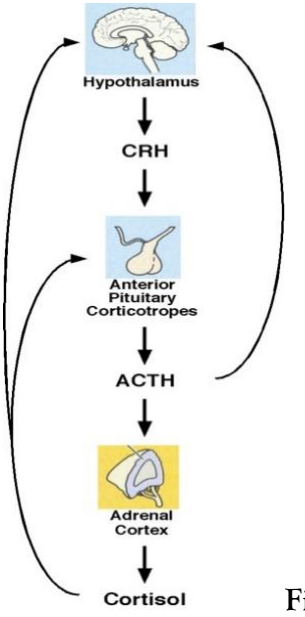


Figure 5

2. Production and release of CRH by the hypothalamus (and subsequent production and release of ACTH and adrenal hormones) is closely affected by many inputs including stress, infection, pain, hypoglycemia, trauma, hemorrhage, and the sleep cycle.

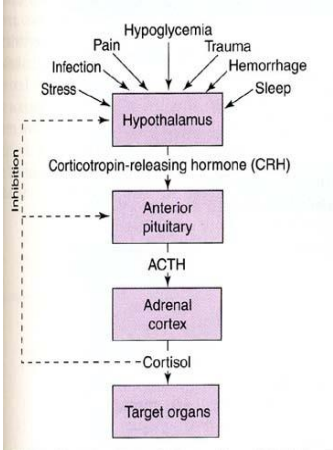


Figure 6

3. Circadian patterns play a major role in the H-P-adrenal axis. The most effective stimulus for H-P-adrenal stimulation is the circadian rhythm. CRH is generally released at increased levels at or around the time of awakening. The circadian pattern involves a 24-hour cyclic pattern that is regulated in part by melatonin and tied to light and dark cycles. This pattern can readily be disrupted by shift work, jet lag, or consistent loss of day and night cycles as may occur in the intensive care unit.

4. For the H-P-adrenal axis, peak hormone levels generally occur between 5 am and 9 am. Lowest levels are generally present in the overnight hours. This plays a role in hypoglycemia that may be seen overnight in hospitalized patients with diabetes receiving exogenous insulin. By convention, cortisol levels are measured at 8 am, presumably at the peak.

5. Even when neurotransmission and circadian patterns are normal, stress stimuli can override the normal pattern. Inputs such as anxiety, pain, illness, surgery, chemical intoxication, and metabolic causes (such as hypoglycemia) may alter H-P-adrenal axis activity.

6. ACTH is produced and released by the corticotropes as part of a complex molecule called pro-opiomelanocortin. POMC includes melanocyte stimulating hormone. Thus, when negative feedback to the anterior pituitary is lost, as with autoimmune destruction of the adrenal cortex in Addison's disease, increased production of POMC (with its components ACTH and MSH) may causes discernable hyperpigmentation of the skin.

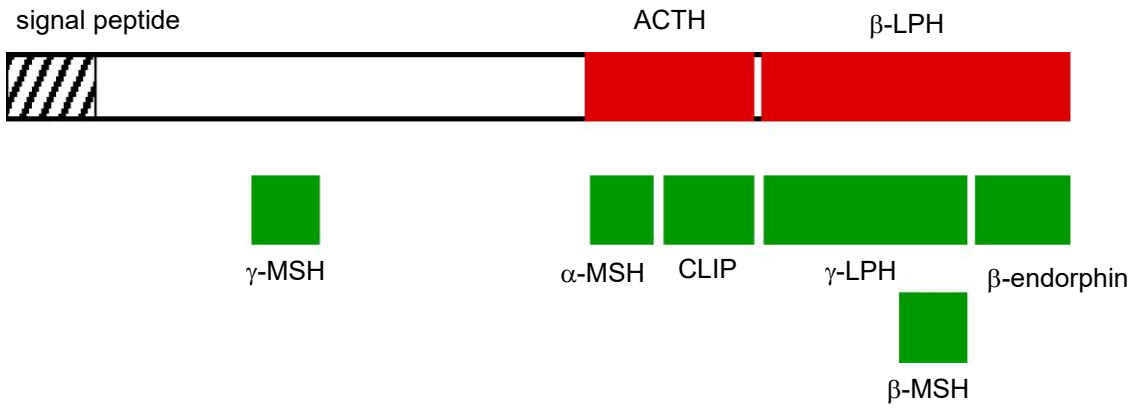


Figure 7

7. ACTH is a 39-amino acid molecule, with the first 18 amino acids essential for its biological activity. Its secretion is episodic and circadian, with blood levels of 150 pg/mL around 8 am, and levels from 0-35 pg/mL at midnight. Its highly variable secretion makes clinical measurement of ACTH levels very challenging. ACTH targets cells of the adrenal cortex with appropriate receptors, particularly the cells of the zona fasciculata and zona reticularis.

8. At the adrenal cortex, ACTH promotes maintenance of adrenal gland size, activation of cAMP, mobilization of cholesterol (the precursor for adrenal hormones), conversion of cholesterol to pregnenolone, and the synthesis and secretion of steroid hormones.

9. ACTH directs synthesis and release of cortisol and androgens and has an indirect effect on the production of the mineralocorticoid aldosterone.

10. Pathophysiology and pharmacology

    a. Excess of cortisol. Can either by primary or secondary. Secondary hyperadrenalism, caused by an adenoma of corticotropes, is termed Cushing's Disease. Primary therapy consists of **transsphenoidal surgery** or **pituitary irradiation**. However, when surgery is delayed, contraindicated, or unsuccessful, medical therapy is often required.

       The somatostatin analog **pasireotide** binds to somatostatin receptors and blocks the release of corticotropin (ACTH) from the corticotropes via its high affinity for somatostatin receptor subtype five. Pasireotide can reduce and normalize urinary free cortisol and improve clinical symptoms. Side effects are similar to other somatostatin analogues and include

gastrointestinal complaints (diarrhea, abdominal pain, loose stools, nausea, flatulence, cholelithiasis, gallbladder sludge/stones, constipation, vomiting, biliary obstruction, anorexia, abdominal cramps). Pasireotide can also cause hyperglycemia and in some cases necessitate the initiation of glucose lowering medications.

Ketoconazole inhibits the first step in cortisol biosynthesis (side-chain cleavage) and, to a lesser extent, the conversion of 11-deoxycortisol to cortisol. The side effects of ketoconazole include headache, sedation, nausea, and vomiting. In men, inhibition of androgen production may lead to gynecomastia, decreased libido, and impotence.

Metyrapone is an 11-beta-hydroxylase (P450c11) inhibitor that blocks the final step in cortisol biosynthesis.

b. Deficiency of cortisol. The first step in diagnosing adrenal insufficiency is to document cortisol deficiency. Low AM cortisol of < 3 mcg/dL is suggestive of diagnosis. An AM cortisol greater than 11 mcg/dL makes clinically significant cortisol deficiency unlikely.

The pharmacologic agent most commonly used to diagnose this is to use **Cosyntropin** which is synthetic ACTH. This has the full biologic potency of native ACTH. This test consists of measuring serum cortisol immediately before and 30 and 60 minutes after intravenous (IV) or intramuscular (IM) injection of 250 mcg of Cosyntropin. The current criteria used to indicate normal adrenal function are a minimum serum cortisol concentration ≥18 mcg/dL before or after Cosyntropin injection. The second step is to measure ACTH. High in primary adrenal insufficiency, low in secondary or tertiary adrenal insufficiency.

Treatment for secondary adrenal insufficiency is **glucocorticoid** administration. In general, there is no need for mineralocorticoid (fludrocortisone) replacement since this is under control of the renin angiotensin system.

F. Structure of the H-P-thyroid axis

1. Thyrotropin releasing hormone (TRH) is released by the hypothalamus. This bathes the thyrotropes of the anterior pituitary and leads to the release of thyroid stimulating hormone (TSH; also known as thyrotropin) TSH enters the general circulation and binds to extracellular receptors on the follicular cells of the thyroid gland. In response, the follicular cells of the thyroid produce and release the thyroid hormones thyroxine (T4) and triiodothyronine (T3).

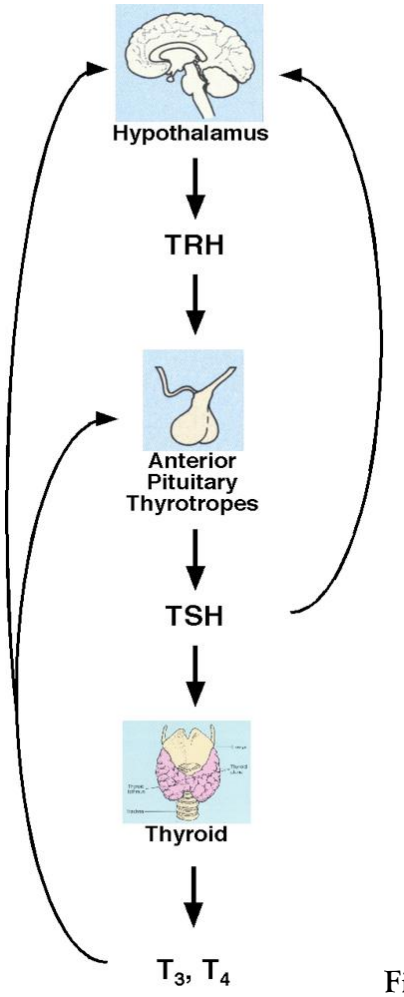


Figure 7a

2. TRH induces the quick release of pre-formed TSH stored in vesicles of the thyrotropes. TRH also induces synthesis of new TSH. The effect of TRH peaks in approximately 10 to 20 minutes.

3. TSH secretion is episodic and pulsatile and displays a circadian rhythmicity. The onset of sleep induces secretion of TSH.

4. The relatively long half-life of TSH allows for effective laboratory measurement at any time of day. However, the presence of stress or illness may alter TSH levels.

5. TSH binds to extracellular receptors on the follicular cells of the thyroid and activates via cAMP. TSH increases vascular flow through the thyroid gland, increases the uptake of iodine, increases production of thyroid hormones and release of T3 and T4 into the general circulation. TSH also acts as a trophic factor, and chronic stimulation with TSH will often lead to cellular hyperplasia and resultant goiter.

6. Pathophysiology and pharmacology

   a. Excess of thyroid hormone. Thyroid-stimulating hormone (TSH)-secreting pituitary adenomas are a rare cause of hyperthyroidism. They account for 0.5 to 3 percent of all functioning pituitary tumors and much less than 1 percent of all cases of hyperthyroidism.
   Labs generally show elevated TSH with an elevated free T4 and or free T3.

   Treatment is **surgery**. Medical therapy is used to treat patients who are not candidates for surgery. It can also be used to restore euthyroidism prior to surgery or for patients that have residual disease after surgery. For medical therapy, most experts recommend a long-acting somatostatin analog **(octreotide)**.

   b. Deficiency of thyroid hormone. Central (secondary/tertiary) hypothyroidism refers to thyroid hormone deficiency due to a disorder of the pituitary, hypothalamus, or hypothalamic-pituitary portal circulation, resulting in diminished thyroid-stimulating hormone (TSH), thyrotropin-releasing hormone (TRH), or both.

   In patients with central hypothyroidism, serum free T4 is low, and serum TSH is usually low as well. In contrast, patients with primary hypothyroidism have an elevated TSH with a low free T4. The treatment of choice for correction of central hypothyroidism is synthetic thyroxine (T4) known as **levothyroxine**. Start with a weight-based levothyroxine dose; start with lower doses in the elderly or in patients with underlying cardiovascular disease. Titrate dose to achieve a normal free T4. (With secondary/tertiary hypothyroidism, cannot use TSH to monitor progress to treatment.)

G. Structure of the H-P-gonadal axis – testicular

1. Gonadotropin releasing hormone (GnRH) is released by the hypothalamus. This bathes the gonadotrophs of the anterior pituitary and leads to the release of follicle stimulating hormone (FSH) and luteinizing hormone (LH). FSH and LH enter the general circulation and bind to extracellular receptors on the Leydig and Sertoli cells of the testes. In response, the Leydig cells produce androgens and the Sertoli cells promote spermatogenesis.

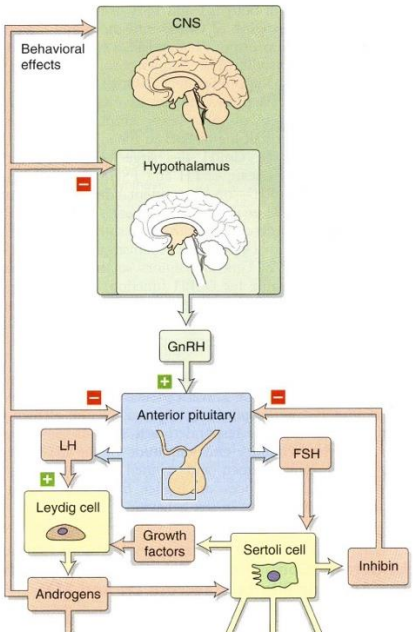


Figure 8

2. Sertoli cells produce inhibin in response to FSH stimulation. Inhibin feeds back in a negative fashion on the anterior pituitary to decrease subsequent release of LH and FSH.

3. Androgens produced by Leydig cells feed back in a negative fashion on the hypothalamus and anterior pituitary to decrease subsequent release of LH and FSH.

4. Pathophysiology and pharmacology: covered in other lectures.

H. Structure of the H-P-gonadal axis – ovarian

1. Gonadotropin releasing hormone (GnRH) is released by the hypothalamus. This bathes the gonadotrophs of the anterior pituitary and leads to the release of follicle stimulating hormone (FSH) and luteinizing hormone (LH). FSH and LH enter the general circulation and bind to extracellular receptors on cells of the ovary. LH binds to thecal cells, producing progesterone and androstenedione. FSH binds to granulosa cells, stimulating these cells to convert androgens, produced by thecal cells, to estradiol by the action of aromatase.

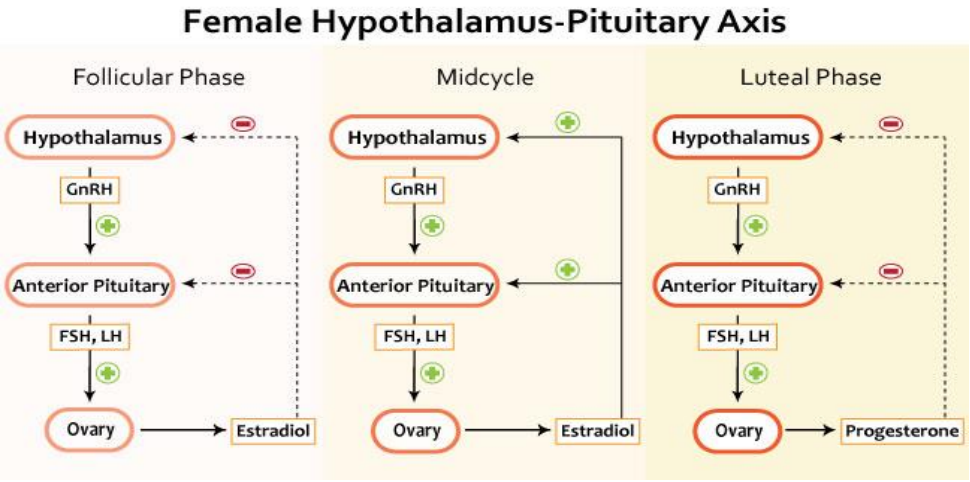


Figure 9

2. In the follicular phase, estradiol feeds back on the hypothalamus and anterior pituitary in a negative fashion to decrease production of GnRH, FSH, and LH.

3. In midcycle, estradiol feeds back on the hypothalamus and anterior pituitary in a positive fashion to increase production of GnRH, FSH, and LH.

4. In the luteal phase, progesterone feeds back on the hypothalamus and anterior pituitary in a negative fashion to decrease production of GnRH, FSH, and LH.

5. The pattern of GnRH changes during the menstrual cycle. During the luteal phase to follicular transition, with low estradiol and low inhibin, GnRH pulses occur every 90-120 minutes and FSH predominates.

6. During mid- to late-follicular phase, with increasing estradiol and inhibin A, GnRH pulses occur more rapidly, and LH predominates.

7. Post-ovulation, with increased estradiol, inhibin A, and progesterone, GnRH pulsatility decreases, and FSH release decreases.

8. Pathophysiology and pharmacology: covered in other lectures.

I. Structure of H-P-breast axis

1. Prolactin is synthesized and secreted by lactotrophs. A prolactin-releasing hormone is hypothesized but has not been definitively identified. Instead, prolactin release is predominantly under tonic inhibition by dopamine from hypothalamic dopaminergic neurons. In the absence of dopamine, prolactin production is uninhibited.

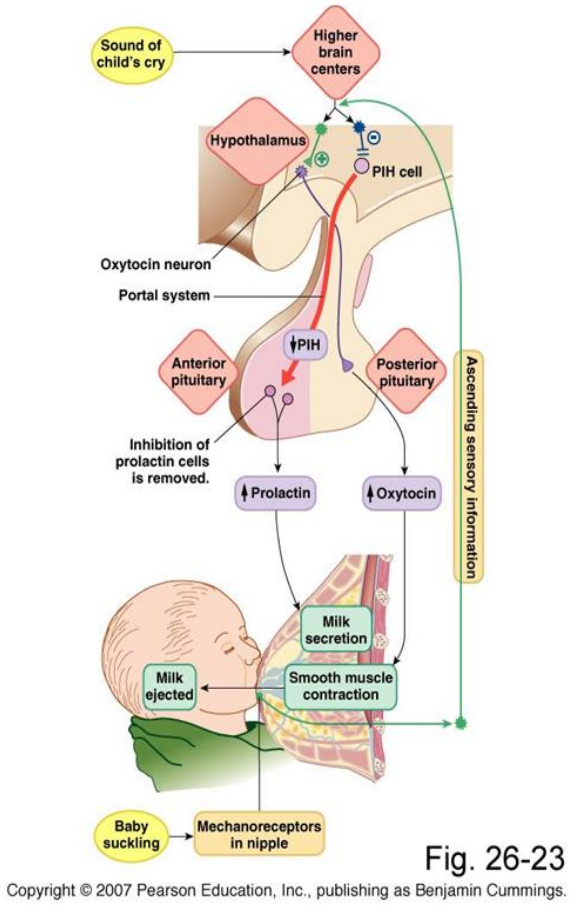


Figure 10

2. Agents that block dopamine receptors, such as typical antipsychotics, may cause increased prolactin levels and resultant adverse effects including galactorrhea, infertility, amenorrhea, and erectile dysfunction.

3. GnRH likely plays some role in prolactin release.

4. Prolactin release is increased by suckling. Suckling inhibits dopamine release, removing inhibition from the lactotroph.

5. Lactotrophs are 15-20% of cells of anterior pituitary; however, this percentage increases with estrogen release, particularly with pregnancy.

6. Pathophysiology and pharmacology.

    a. Excess of prolactin. Prolactinoma is treated with medical management even in the presence of abnormal visual fields. Drugs may cause shrinkage of the tumor and sometimes they can be stopped after 1-2 years of treatment.

Bromocriptine and **cabergoline** are ergot derivatives with a high affinity for dopamine D2 receptors. They are known as dopamine agonists. Dopamine agonists can cause nausea, headache, light-headedness, orthostatic hypotension, and fatigue.

b. Deficiency of prolactin. The only known clinical manifestation of prolactin deficiency is the inability to lactate after delivery. Isolated prolactin deficiency is rare; most patients with acquired prolactin deficiency have evidence of other pituitary hormone deficiencies. There is no specific replacement.

J. Hormones of the posterior pituitary

1. Extension of the hypothalamus – contains axon terminals of neurons located in supraoptic and paraventricular nuclei.

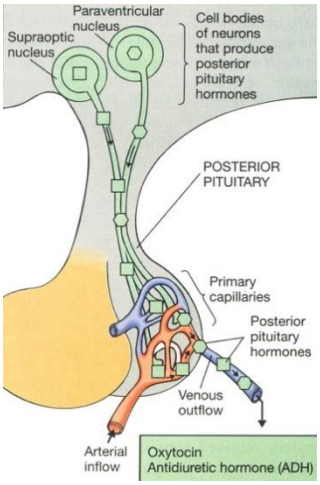


Figure 11

2. Neurons generate action potentials – produce membrane depolarization and exocytosis of contents of secretory vesicles in a neuroendocrine manner.

3. Oxytocin and anti-diuretic hormone (ADH) enters the systemic circulation through venous drainage of the posterior pituitary into the intracavernous sinus and internal jugular vein.

K. Anti-diuretic hormone

1. ADH is a peptide hormone released by the posterior pituitary in response to rising plasma tonicity or falling blood pressure. It possesses antidiuretic and vasopressor properties. A deficiency of this hormone results in diabetes insipidus.
2. ADH (also known as arginine vasopressin, or AVP) binds with the V_2 receptors on the cells of the collecting duct, initiating the insertion of the Aquaporin 2 water channels. Without ADH, these channels are stored in endosomes within the cytoplasm.
3. With ADH, the cells of the collecting duct become permeable to water, water is reabsorbed into medullary interstitium and eventually into blood, and urine becomes more concentrated as water is reabsorbed.
4. The most important stimuli for ADH release are an increase in plasma osmolality and decrease in blood volume.
5. Pathophysiology and pharmacology.
    a. Excess of ADH. Syndrome of inappropriate anti-diuretic hormone. Treat underlying causes. Discussed in Block 4A.
    b. Deficiency of ADH. Central diabetes insipidus (CDI) is characterized by decreased release of antidiuretic hormone (ADH), resulting in a variable degree of polyuria. Lack of ADH can be caused by disorders that act at one or more of the sites involved in ADH secretion.

        Desmopressin (dDAVP) is a two-amino acid substitute of ADH that has potent antidiuretic but no vasopressor activity. Acts on V2 receptors in the distal tubules and collecting ducts (cAMP) increased water permeability. Available as an intranasal spray, tablet, and parenteral formulation for subcutaneous injection. Most important side effect is hyponatremia. Hyponatremia can usually be avoided by giving the minimum desmopressin dose that is required to control the polyuria. The initial aim of therapy is to reduce nocturia, thereby permitting adequate sleep. Thus, the first dose is typically given at bedtime. Follow serum sodium.

L. Oxytocin

1. The release of oxytocin is stimulated during lactation and childbirth. In the lactating breast, oxytocin stimulates milk ejection by stimulating contraction of cells lining alveoli and ducts in the mammary gland.
2. In the pregnant uterus, oxytocin produces rhythmic contractions to help induce labor and promote uterine regression after delivery. At the onset of labor, there is upregulation of oxytocin receptors in uterine muscle – up to 200 times that of the non-pregnant uterus. Contractile activity of the uterus acts through positive feedback during parturition to stimulate oxytocin neurons.
3. Oxytocin may play a role in maternal behavior and amnesia in childbirth.
4. Increased levels of oxytocin are noted during moments of intimacy. This may breed trust and help form that most mysterious of human concepts – love.
5. Pharmacology. Synthesized oxytocin is administered intravenously for initiation and augmentation of labor (**Pitocin**; you will hear it referred to as "pit" on the labor and delivery floor.)

Answers to Study Questions

1. How do end-organ hormones feed back on the components of the hypothalamic-pituitary-end organ axis in most cases?

   End organ hormones negatively feed back to the hypothalamus, the pituitary, and to the end organ itself, mitigating production and release of their particular hormones.

2. What is unique about growth hormone when compared to all other hormones produced by the anterior pituitary?

   Growth hormone is the only anterior pituitary hormone with significant effects systemically and on a specific end organ.

3. Why does ACTH overproduction, when seen in Addison's disease, lead to skin hyperpigmentation?

   ACTH is produced and released by the corticotropes as part of a complex molecule called pro-opiomelanocortin. POMC includes melanocyte stimulating hormone. Thus, when negative feedback to the anterior pituitary is lost, as with autoimmune destruction of the adrenal cortex in Addison's disease, increased production of POMC (with its components ACTH and MSH) may cause discernable hyperpigmentation of the skin.

4. Why are serum TSH measurements generally reliable, but serum ACTH measurements often are not?

   ACTH secretion is episodic and circadian, with blood levels of 150 pg/mL around 8 am, and levels from 0-35 pg/mL at midnight. Its highly variable secretion makes clinical measurement of ACTH levels very challenging. In contrast, although TSH release is pulsatile, the relatively long half-life of TSH allows for effective laboratory measurement at any time of day. However, the presence of stress or illness may alter TSH levels.

5. What is the product of Sertoli cells? How does that product lead to negative feedback of the H-P-gonadal axis in persons with testes?

   Sertoli cells produce inhibin in response to FSH stimulation. Inhibin feeds back in a negative fashion on the anterior pituitary to decrease subsequent release of LH and FSH.

6. What are the main components of the H-P-gonadal axis in persons with ovaries? How does this axis function?

   Gonadotropin releasing hormone (GnRH) is released by the hypothalamus. This bathes the gonadotrophs of the anterior pituitary and leads to the release of follicle stimulating hormone (FSH) and luteinizing hormone (LH). FSH and LH enter the general circulation and bind to extracellular receptors on cells of the ovary. LH binds to thecal cells, producing progesterone and androstenedione. FSH binds to granulosa cells, stimulating these cells to convert androgens, produced by thecal cells, to estradiol by the action of aromatase.

7. What is unique about the H-P-breast axis when compared to other H-P-end organ axes?

   Prolactin is synthesized and secreted by lactotrophs. A prolactin-releasing hormone is hypothesized but has not been definitively identified. Instead, prolactin release is predominantly under tonic inhibition by dopamine from hypothalamic dopaminergic neurons. Other H-P-end organ axes have definitive stimulator hormones from the hypothalamus.

8. How does oxytocin act and why does it become so potent at the time of labor?

   In the pregnant uterus, oxytocin produces rhythmic contractions to help induce labor and promote uterine regression after delivery. At the onset of labor, there is upregulation of oxytocin receptors in uterine muscle – up to 200 times that of the non-pregnant uterus. Contractile activity of the uterus acts through positive feedback during parturition to stimulate oxytocin neurons.

9. a

10. d

11. a

12. d

13. b